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Prediction of Near-Surface Velocity from Seismic Traveltimes Data Using Machine Learning

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Abstract

Accurate determination of near-surface velocity is essential for calculating static corrections for seismic data, as neglecting shallow velocity can lead to significant misplacement of deeper seismic reflectors. However, traditional methods for inverting refracted traveltimes data are often computationally intensive and time-consuming. To overcome these limitations, we present a novel machine learning-based inversion framework that predicts the thickness and velocity of near-surface layers from refracted traveltimes data. Our approach utilizes a data-driven methodology to approximate the relationship between seismic traveltimes and corresponding velocity and thickness, reducing ambiguities in seismic inversion results, particularly without sufficient prior information. We trained a deep neural network (DNN) model using synthetic traveltimes datasets and evaluated its performance on synthetic and real-world data. The results demonstrate that the proposed approach can accurately estimate layer thicknesses and velocities with high consistency between predicted and actual values. Furthermore, applying the machine learning approach to a field dataset acquired over a dune in Saudi Arabia, comprising 37 receivers at 2 m intervals, yielded results consistent with borehole data. The proposed machine learning framework offers several advantages over traditional inversion techniques, including automatically identifying valuable inversion information without prior knowledge, avoiding non-uniqueness challenges, and efficient training on smaller datasets without extensive computational resources. Additionally, the approach can manage nonlinearity associated with seismic inversion and reduce dependence on initial models, providing automatic solutions in milliseconds. Although this approach depends on the quality and amount of training data, it marks a notable step forward in seismic inversion and offers substantial potential to enhance the accuracy and efficiency of static seismic corrections.

Introduction

Near-surface seismic velocity is a critical parameter in seismic exploration, as it directly affects the accuracy of subsurface imaging and reservoir characterization. Traditional methods for estimating seismic velocity from traveltimes data are often time-consuming, labor-intensive, and prone to human error. Since the

early days of seismic exploration, various techniques have been developed for estimating seismic velocity, including tomographic inversion and refraction analysis. These methods, such as the ray-based approach (Aki and Richards, 1980) and the wavefront reconstruction method (Schneider, 1998), have shown promise in estimating seismic velocity. However, they are often noise-sensitive and require careful data processing and interpretation. Moreover, these methods can be computationally intensive and rely on simplifying assumptions. Therefore, there is still a need for a fast, automatic, and robust technique for estimating seismic velocity, particularly in the presence of complex subsurface structures and noisy data. An emerging solution lies in machine learning, which excels at identifying complex patterns in seismic data and delivering precise velocity estimates.

Machine learning algorithms, such as deep learning neural networks (DNNs) and convolutional neural networks (CNN), are highly effective in solving classification, regression, and optimization problems (LeCun et al., 2015; Goodfellow et al., 2016). These techniques have gained significant attention in geophysics in recent years due to their ability to learn complex patterns in large datasets and make accurate predictions (Bishop, 2006; Jordan and Mitchell, 2015). The application of machine learning in geophysics has been further facilitated by the development of open-source libraries such as TensorFlow and PyTorch, which provide efficient and scalable implementations of various machine-learning algorithms (Abadi et al., 2016; Paszke et al., 2019).

DNNs have become increasingly popular in recent years (LeCun et al., 2015; Goodfellow et al., 2016). Supervised DNNs are trained on labeled datasets where the input-output mapping is explicitly defined, allowing the network to learn the relationship between the inputs and outputs (Krizhevsky et al., 2012). On the other hand, unsupervised DNNs do not require labeled data and instead learn to identify patterns and relationships in the data through self-supervised learning (Vincent et al., 2010; Goodfellow et al., 2016).

This study uses first arrival traveltimes data to explore the effectiveness of a supervised DNN in estimating the thickness and seismic velocity of near-surface layers. The network is trained on synthetic seismic responses from a range of near-surface velocity models. Its performance is evaluated using unseen synthetic examples and field data collected over a dune. The DNN is designed to predict key parameters, including the velocity of the weathered layer, the depth to bedrock, and the bedrock velocity. The overarching objective is to develop a reliable and precise method for estimating near-surface seismic velocity, which is an essential factor in seismic exploration and geotechnical engineering.

Deep Neural Network (DNN)

The concept of DNNs has advanced considerably since their origins in early supervised neural networks, which were initially modeled after the brain's neural processes (Aki and Richards, 1980; Schneider, 1998). This field has seen remarkable progress over the years, propelled by innovations in training algorithms, network architectures, and computational techniques (Abadi et al., 2016; Paszke et al., 2019). Modern DNNs (Figure 1) are composed of multiple layers of interconnected nodes or neurons, which are arranged in a hierarchical structure to learn complex patterns in data (Bengio et al., 2013; Goodfellow et al., 2016). These networks typically consist of an input layer, one or more hidden layers, and an output layer, with each layer connected to the next through weighted connections (Krizhevsky et al., 2012). In a DNN, each neuron or processing element (PE) receives input signals from the previous layer, applies a nonlinear transformation to the weighted sum of these inputs, and then sends the output to the next layer (LeCun et al., 2015; Vincent et al., 2010). The most common activation functions used in DNNs include the rectified linear unit (ReLU), sigmoid, and tanh functions (Glorot et al., 2011; Nair et al., 2010). Using multiple hidden layers allows DNNs to learn complex and abstract data representations, solving challenging problems in image and speech recognition, natural language processing, and other areas (Bengio et al., 2013; Goodfellow et al., 2016). In this study, we use the DNN model to estimate the thickness of near-surface layers from traveltimes data, which offers a promising alternative to traditional methods. The proposed methodology can potentially

improve the accuracy and efficiency of near-surface characterization, while reducing costs and enhancing collaboration between geoscientists and engineers.

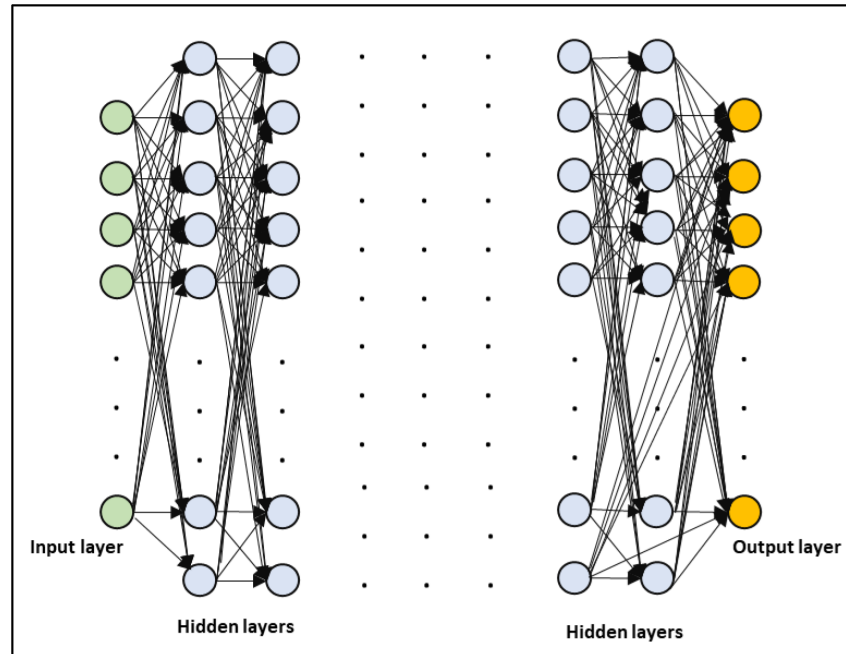


Figure 1—Deep Neural Network (DNN) structure.

Implementation

In this study, we employ a two-layer model to characterize the subsurface, where the thickness of the near-surface layer is estimated using a DNN trained on traveltimes data. The DNN is designed to learn the complex relationship between the traveltimes measurements and the thickness of the near-surface layer, allowing for accurate and efficient estimation of this parameter. Once the thickness of the near-surface layer is estimated, we utilize a classical inversion approach to estimate the velocities of the two layers. This hybrid approach leveraged the strengths of machine learning and traditional geophysical inversion methods, enabling us to characterize the subsurface comprehensively. The classical inversion method, which relied on the finite-difference solution of the Eikonal equation and a conjugate-gradient optimization algorithm, is used to estimate the velocities of the two layers, given the estimated thickness of the near-surface layer. This implementation enabled us to leverage the strengths of both methods, leading to a reliable and precise estimation of subsurface parameters, including the near-surface layer thickness and the velocities of the upper and lower layers.

Our model collects traveltimes data at 24 receivers from just two shot locations, one at each end of the profile. Training an effective neural network requires a sufficiently large and diverse dataset. For comparison, [Araya-Polo et al. \(2018\)](#) generated thousands of randomized 2D velocity models for DNN training, while [Yang and Ma \(2019\)](#) used 1,730 velocity models for seismic data simulations. In designing our dataset, we divided the subsurface into 768 rectangular cells (48 rows $\times$ 16 columns), each measuring 0.5 m by 0.5 m. Observation points were spaced every meter for the forward modeling. Since supervised neural networks typically require preprocessing, such as standardizing, the input data were normalized accordingly. The DNN was trained using profile data from the two shots to estimate the thickness of the near-surface layer beneath each receiver. A total of 691,200 model samples were generated through forward simulations, covering a wide range of upper- and lower-layer velocities to ensure sufficient variability for training.

The advent of modern DNN architecture has enabled the estimation of complex subsurface parameters, such as seismic velocities, with unprecedented accuracy. In this study, we employ a state-of-the-art DNN to

estimate the seismic velocities of the upper and lower layers, leveraging the strengths of advanced neural network architectures, including convolutional and recurrent layers. The proposed DNN structure consists of a sequential arrangement of six layers, carefully designed to facilitate effective learning and prediction. The network begins with a normalization layer, standardizing the input data to have a consistent scale, resulting in an output shape of (46) and requiring 93 parameters. This is followed by a series of dense (fully connected) layers, with the first layer comprising 64 neurons and 3,008 parameters, the second dense layer having 128 neurons and 8,320 parameters, and subsequent layers having 64, 32, and 24 neurons with 8,256, 2,080, and 792 parameters, respectively. Notably, the last dense layer has been modified to include L2 regularization, which adds a penalty term to the loss function to discourage large weights and prevent overfitting. The L2 regularization term helps reduce the model's complexity and improve its generalization capabilities, making it more robust and reliable for predicting continuous outcomes. The gradual decrease in the number of neurons across the dense layers, combined with the regularization technique, allows the network to progressively refine its feature extraction and representation capabilities, yielding a compact and informative output. Overall, the modified DNN structure is well-suited for learning and predicting continuous outcomes, making it an effective tool for solving regression problems while mitigating the risk of overfitting.

Our DNN was trained on a comprehensive dataset, carefully tuning hyperparameters, including a learning rate of 0.001, an Adam optimizer, and a ReLU activation function. The trained model was then used to estimate the velocities of the upper and lower layers using classical non-linear inversions, which were subsequently compared to the actual values. This hybrid approach, combining the strengths of machine learning and traditional geophysical inversion methods, enables robust and accurate estimation of subsurface parameters. The DNN was trained on a subset of 80% of the available examples, with the remaining 20% reserved for validation purposes. The training process was conducted over 500 epochs, allowing the DNN to learn the complex relationships between the input data and the target outputs. During training, the DNN's performance was evaluated using the mean squared error (MSE) loss function, which measured the average squared difference between the predicted and actual values (Figure 2). The training loss was monitored at each epoch, and the validation loss was evaluated at regular intervals to assess the DNN's generalization performance. The validation loss was also calculated using the MSE loss function, which quantitatively measures the DNN's ability to predict the target outputs on unseen data. By monitoring both the training and validation losses, we identified the optimal point to stop training, preventing overfitting and ensuring that the DNN generalized well to new, unseen data.

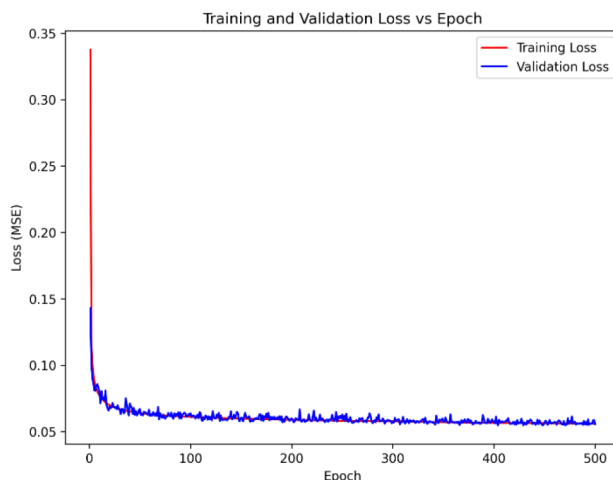


Figure 2—Loss plot from the training process for the model. Following a standard practice of an 80%–20% training/validation split of the total data set.

The DNN was thoroughly evaluated using previously unseen models to assess its predictive performance. Figure 3 shows results from six randomly selected examples of near-surface layer thickness prediction, highlighting the network's impressive accuracy. The predicted values closely match the actual values, indicating strong generalization and consistent reliability, even on data not encountered during training. This robust performance underscores the model's ability to capture key subsurface patterns and relationships, making it a promising tool for real seismic applications. The tight correspondence between predicted and actual values in these test cases reinforces confidence in the DNN's precision and effectiveness for estimating near-surface layer thickness.

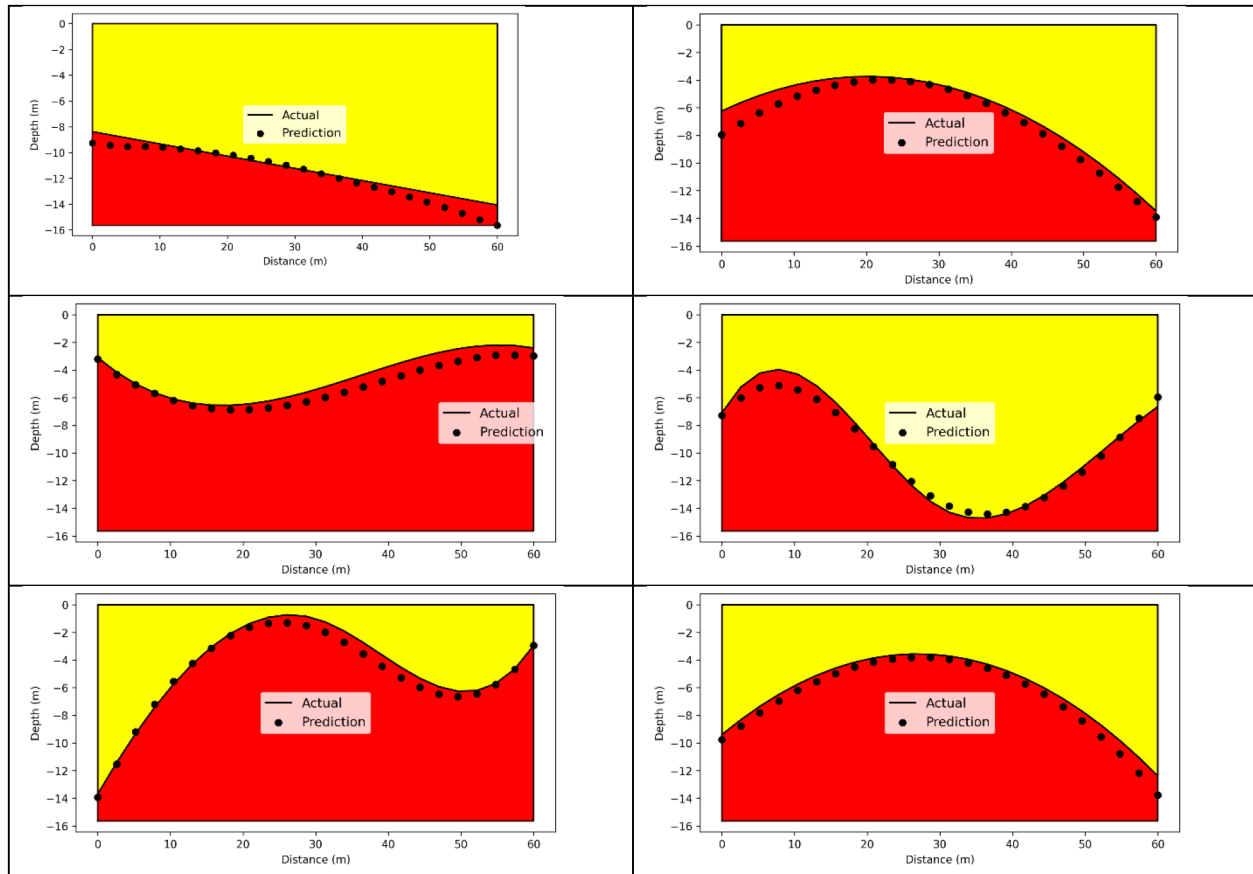


Figure 3—Predicted models (black circles) obtained using the trained DNN from traveltimes data of 6 random and complex models.

Case Study

A comparative analysis was conducted to validate the efficacy of our proposed DNN approach against conventional inversion methods. We applied our approach to a seismic refraction dataset acquired over a barchan dune in Saudi Arabia. We compared the results with those obtained by Hanafy et al. (2020) using a conventional inversion method. The dataset comprised traveltimes that satisfied the reciprocity condition, as Hanafy et al. (2020) outlined. In this experiment, the seismic source consisted of an 8 kg sledgehammer striking a $20 \times 20 \times 4$ cm³ metal plate on the sand surface. A total of 73 shot gathers were recorded at 1-meter intervals, each using 37 receivers spaced 2 meters apart. Traveltimes from the first and last shots were selected, manually picked, and interpolated at equally spaced 24 locations, as shown in Figure 4a. The conventional method used for comparison involved solving the ray equations with a finite-difference approach, followed by a conjugate-gradient inversion that required 20 iterations to achieve convergence (Hanafy et al., 2020). In contrast, our DNN approach yielded a simpler velocity model with only two distinct

layers (Figure 4b). The predicted depths to the first layer beneath each receiver revealed a refractor interface that can be approximated by a layer with a depth of 8 m at the eastern extremity, dipping westward to approximately 10 m. The estimated velocities for the top and lower layers were 366 m/s and 3728 m/s, respectively. Figure 4c presents the traveltimes computed from the DNN-predicted model. For comparison, the conventional inversion approach yielded a more detailed velocity tomogram featuring three distinct layers (Figure 4d). The first layer displayed a velocity gradient increasing from 300 m/s at the surface to 1300 m/s at a depth of 7 meters, with its base gently dipping westward. The second layer showed velocities between 1700 m/s and 2600 m/s, beginning at depths between 7 and 7.25 meters. High velocities (≥ 3000 m/s) characterized the third and deepest layer, representing the fully saturated, mud-rich sabkha zone. The main differences between the two approaches lie in the number of layers, velocity gradients, and depth and geometry, where our DNN approach identified two distinct layers with uniform velocities. In contrast, the conventional inversion method revealed three layers with more complex velocity gradients, particularly in the first layer, and a more complicated and dipping refractor interface. These differences highlight each approach's strengths and limitations, with our DNN method providing a more straightforward and efficient solution. In contrast, the conventional inversion method offers a more detailed and nuanced representation of the subsurface velocity structure. Notably, the output from our DNN approach can be utilized as a valuable initial model for the conventional inversion method, potentially improving its efficiency and accuracy. Using the DNN-predicted velocity model as a starting point, the conventional inversion method can refine and detail the model, leveraging the strengths of both approaches. This hybrid approach can combine the speed and simplicity of the DNN method with the detailed and nuanced representation of the conventional inversion method, ultimately leading to a more accurate and comprehensive understanding of the subsurface velocity structure.

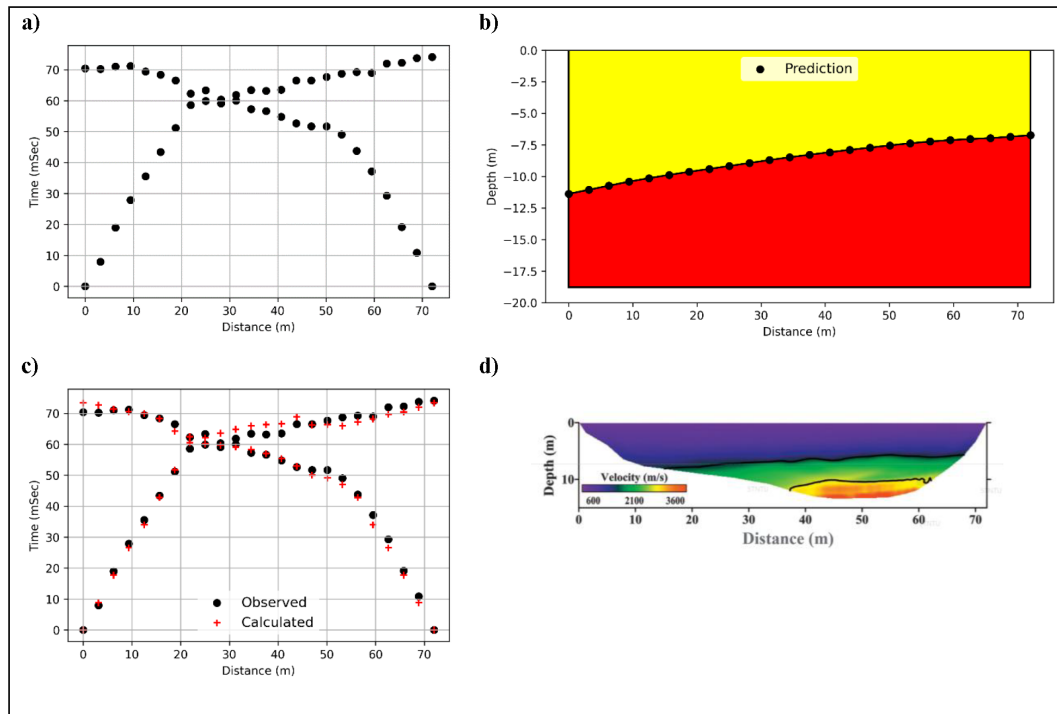


Figure 4—(a) Offset-time curves showing the manually picked traveltimes for shots 1 and 73. (b) The seismic velocity model as derived using the machine learning approach. (c) Traveltimes response computed from the machine learning-predicted velocity model. (d) P-wave velocity tomogram in the depth-offset domain, generated from conventional seismic survey data by Hanafy et al. (2020).

Conclusions

This study has investigated the efficacy of a DNN in estimating thickness and seismic velocity from surface seismic traveltimes data. The DNN was trained on synthetic seismic traveltimes signatures of diverse subsurface models, encompassing various velocity structures and geological formations. The network's performance was evaluated using theoretical and field data, and its results were compared to those obtained from traditional seismic tomography methods. The proposed DNN method was developed and trained using the ReLU activation function, Adam optimizer, and L2 regularization, which enhanced the network's generalization ability. The key findings of this study reveal that DNN is a potent tool for near-surface characterization, capable of accurately estimating thickness and velocity. Notably, DNN exhibits strong learning and generalization abilities, rendering it a valuable reference for future studies. Furthermore, the proposed machine learning approach efficiently mitigates the nonlinearity associated with seismic inversion, reducing dependence on initial velocity models. Unlike conventional seismic inversion algorithms, the proposed framework is automatic, providing rapid and accurate solutions in milliseconds. The proposed approach can potentially improve the accuracy and efficiency of near-surface characterization, and future studies can build upon this work by exploring alternative DNN architectures and techniques further to enhance the accuracy and robustness of near-surface characterization.

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